

EES301: Global biogeochemical cycles and Climate change

The Global Nitrogen Cycle

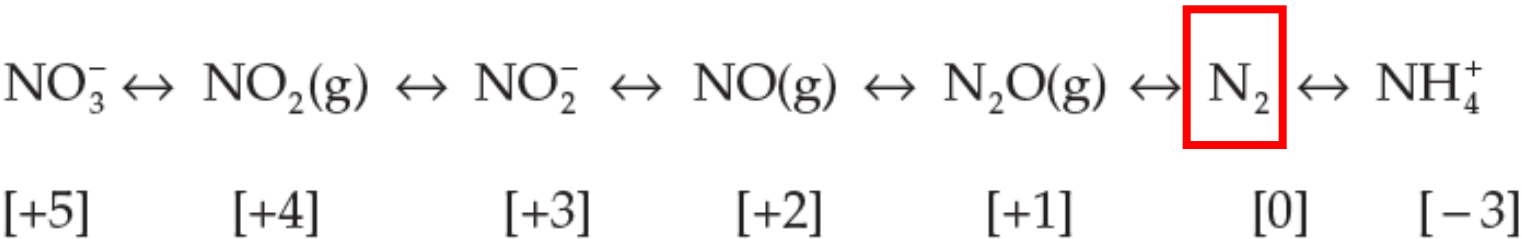
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EES301

Introduction

- “N” is the most common element in Earth's atmosphere, yet it (fixed or available form) is one of the most limiting nutrients for primary production.
- It is an important component of biological macromolecules and atmospheric processes.
- Anthropogenic processes - major pollutants of water bodies and the atmosphere; similar level effects like natural N_2 fixation – substantial changes to the global N cycle.
- Biological transformations contribute majorly to N fluxes across major reservoirs.

**Biogeochemically
important N
compounds and
their oxidation
states**

Compound	Name	N oxidation state
NH ₂ , NH ₄ ⁺	Ammonia, ammonium	−3
R-NH ₃	Amino acids, organic nitrogen polymers	−3
NH ₂ OH	Hydroxylamine	−1
N ₂	Dinitrogen gas	0
N ₂ O	Nitrous oxide	+1
NO	Nitric oxide	+2
HONO, NO ₂ [−]	Nitrous acid, nitrite	+3
NO ₂	Nitrogen dioxide	+4
N ₂ O ₅ , HNO ₃ , NO ₃ [−]	Dinitrogen pentoxide, nitric acid, nitrate	+5



Ward B., 2012,
The Global N cycle

Major Nitrogen reservoirs on Earth

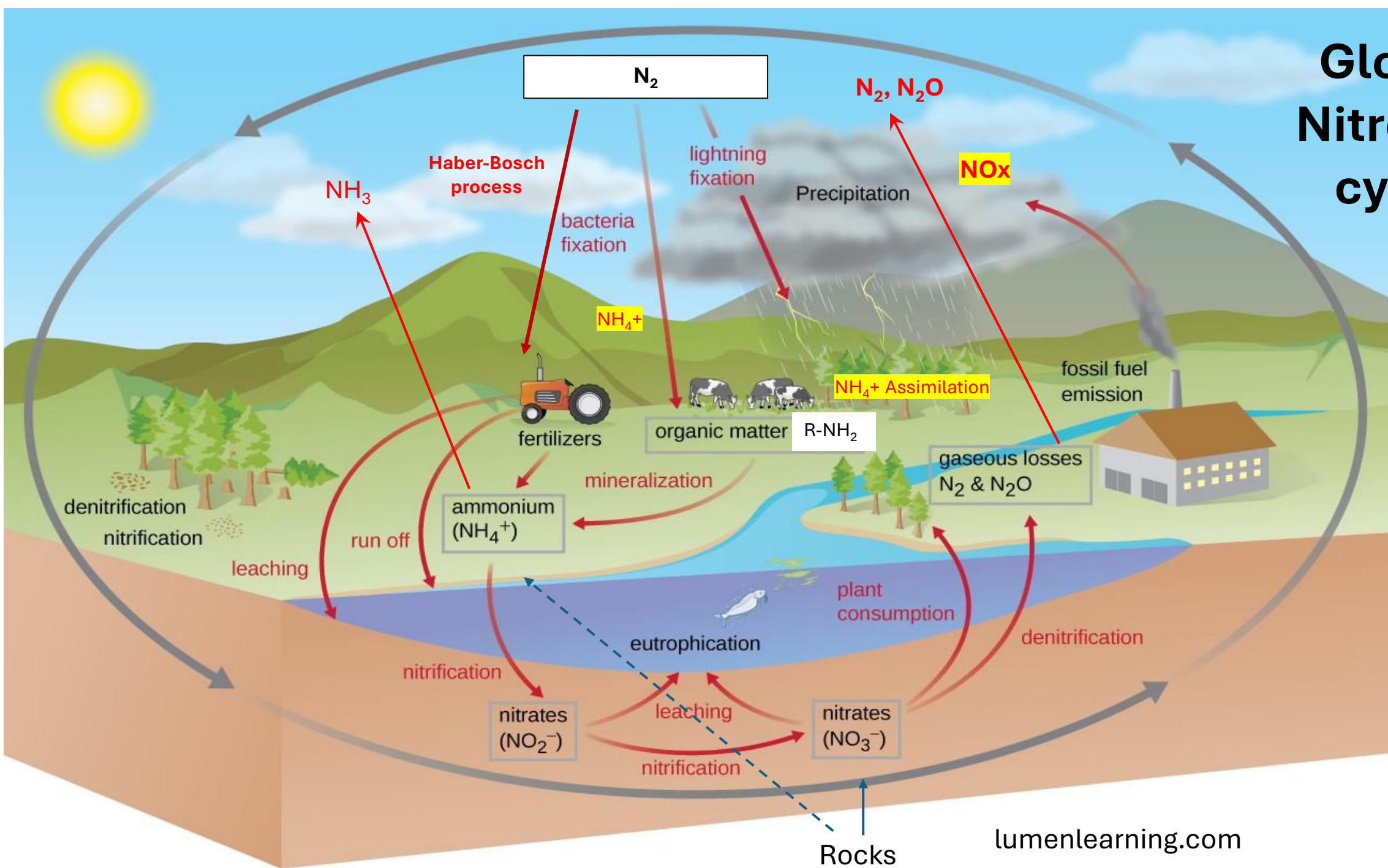
PON/DON:
Particulate/Dissolved
Organic Nitrogen

*1000 Teragrams (Tg) = 1 Gt

Ward B., 2012,
The Global N cycle

Reservoir		Tg *	
Atmosphere	N ₂	3.7 × 10 ⁹	~ 27%
	N ₂ O	1.4 × 10 ³	
Biosphere	Marine	3.0 × 10 ³	~ 0.000062%
		5.0 × 10 ²	
	Terrestrial	5.4 × 10 ³	
		2.9 × 10 ⁴	
Ocean	N ₂	7.7 × 10 ³	
		1.46 × 10 ⁶	
Geological	N ₂ O	0.34	~ 0.005%
	NO ₃ ⁻	6.0 × 10 ⁵	~ 73%
	PON	9.0 × 10 ⁴	
	DON	8.1 × 10 ⁵	
	Continental crust	1.3 × 10 ⁹	
	Crustal rocks	6.4 × 10 ⁸	
	Oceanic crust	8.9 × 10 ⁵	
	Coastal sediments	3.2 × 10 ⁴	
	Deep ocean sediments	2.0 × 10 ⁹	
	Soil	1.4 × 10 ⁵	
		2.2 × 10 ⁴	

Global Nitrogen cycle

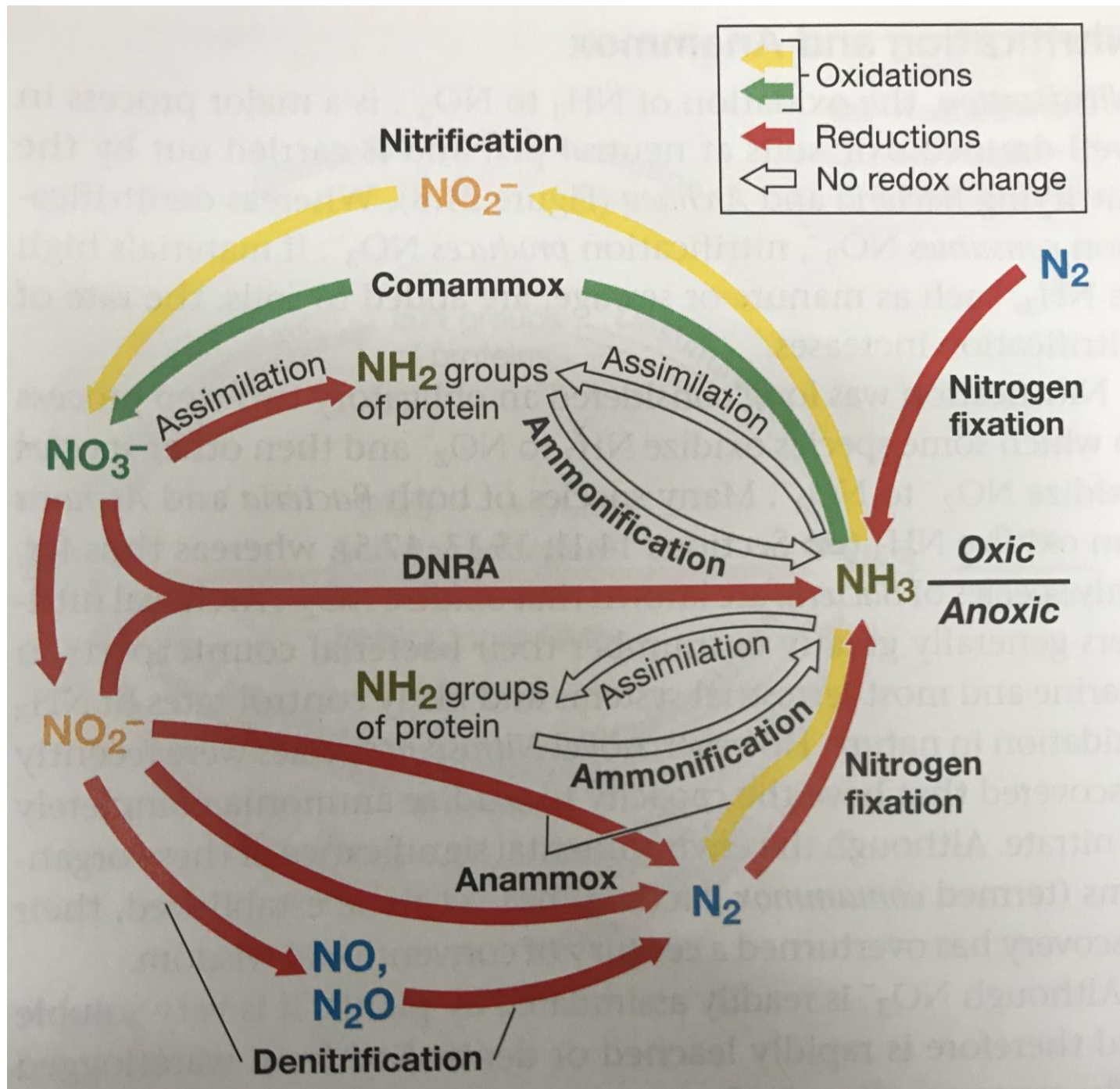


Major processes in active N cycling

- **Biological** – several processes from N_2 fixation, assimilation, decomposition to denitrification
- **Anthropogenic**
 - Haber-Bosch process; intentional production of NH_3
 - Combustion of fossil fuels/biomass; unintentional production of NO_x
 - Industrial agriculture; intensification of biological N_2 fixation
- **Atmospheric**
 - several reactions; **though fluxes are smaller than biological fluxes, these are important for climate, stratospheric ozone and photochemical smog**

E.g., $\text{NO}_2 + h\nu \rightarrow \text{NO} + \text{O}$; $\text{O}_2 + \text{O} \rightarrow \text{O}_3$

$$\text{NO}_2 + \text{OH}^- \rightarrow \text{HNO}_3$$
$$\text{NH}_3 (\text{g}) + \text{HNO}_3 (\text{aq}) \rightarrow \text{NH}_4\text{NO}_3 (\text{s}) \text{ OR } \text{NH}_4^+ (\text{aq}) + \text{NO}_3^- (\text{aq})$$



Biological N cycling processes

**DNRA: Dissimilatory nitrate reduction to ammonium*

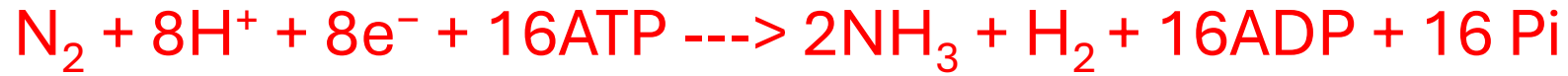
Brock Biology of Microorganisms,
15th edition

Assimilation

- Plants acquire fixed N through:
 - ammonium and/or nitrate fertilizer or manure
 - release of N compounds during organic matter decomposition
 - N_2 fixed to ammonia by microbes
 - N_2 fixed by lightning
- Most microbes utilize NH_4^+ or obtain N by ingestion of amino acids, nucleotides, and other small organic-N molecules; some fix N_2
- Other organisms utilize fixed forms of N

1. Biological N₂ fixation

- Microorganisms fix ≈60% nitrogen required by plants



- Relatively small number of bacteria and archaea can do it
- **Nitrogenase enzyme complex** catalyze the process; sensitive to oxygen.

2. Ammonification/Mineralization

- Decomposition of N-containing organic matter (e.g., proteins) and nitrogenous waste (e.g., urea) into ammonia.



- Majority ammonia released into soils/water bodies - available for assimilation and nitrification; only a small part is released to the atmosphere under certain conditions (e.g., high pH)

3. Nitrification

- Oxidation of ammonia to nitrate (via nitrite) in oxic environments
- Performed by nitrifying bacteria and archaea

- **Two-step oxidation process**

Ammonia oxidation



Nitrite oxidation



- **Comammox process** (Complete Ammonia Oxidation) – single type of microbes catalyze the complete nitrification process (one step)



Fate of NO_3^- :

- Plant uptake
- Leaching/Run-off
- Denitrification

4. Anaerobic ammonia oxidation (Anammox)

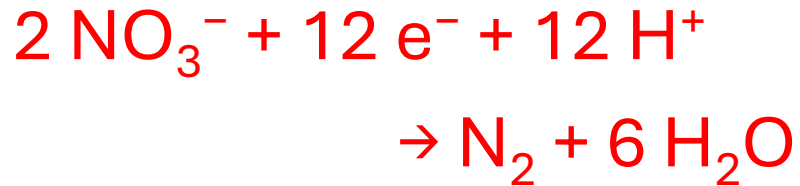
- Ammonia oxidation under anaerobic conditions - with nitrite (or NO) as the electron acceptor, forming N₂ as the product.



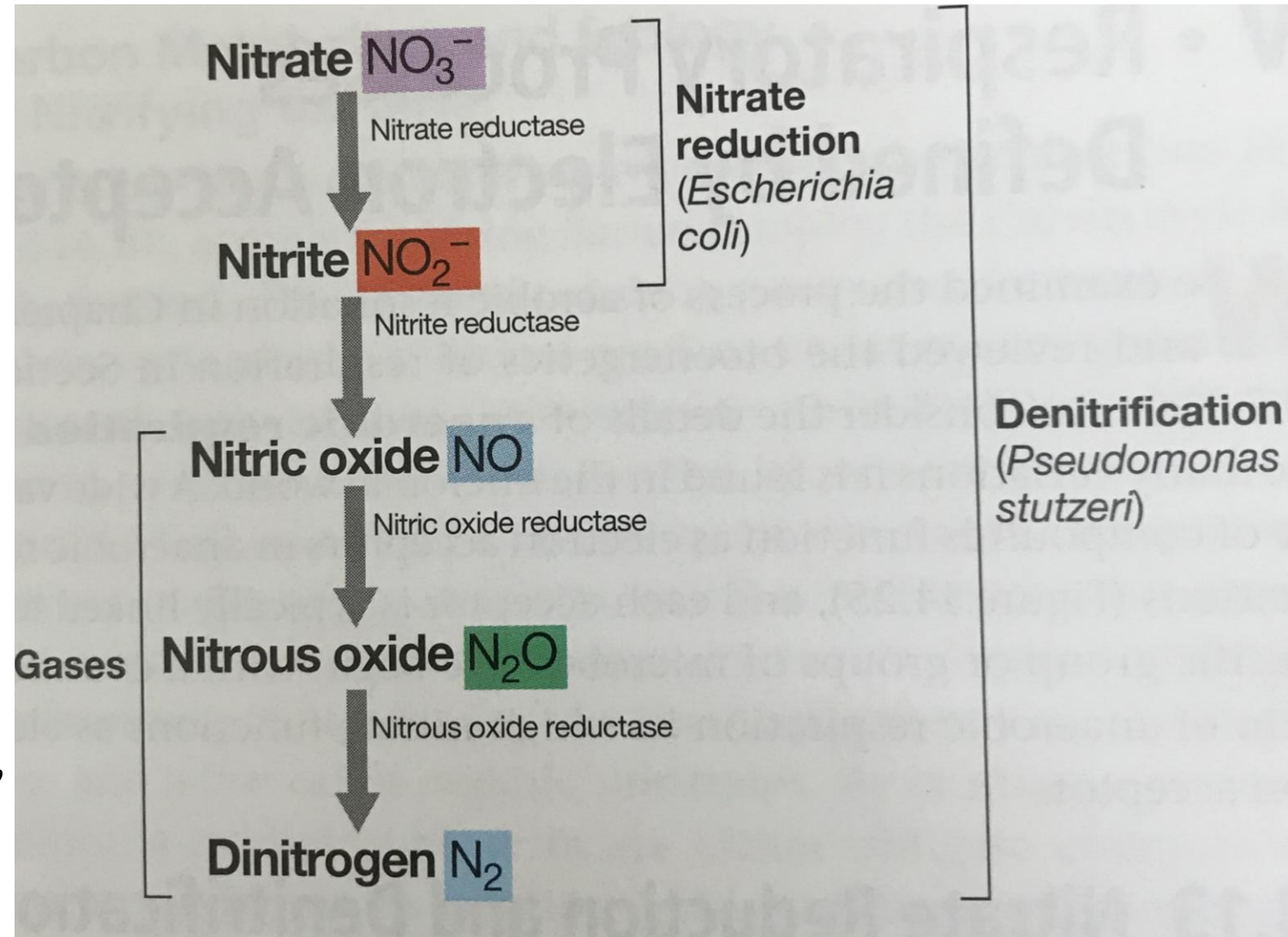
- This process makes up a major proportion of N₂ formation (up to 60 %) in the marine and freshwater ecosystems.
- **Ecological significance** – limits primary productivity in oceans through continuous removed of fixed N forms
- Application in the removal of ammonia in wastewater treatment plants

6. Denitrification

- Nitrate reduction to N_2



- Anaerobic respiration process with NO_3^- as a terminal electron acceptor; common in, e.g., wetlands, anoxic soils, sediments



Brock Biology of Microorganisms, 15th edition

Implications of denitrification

- N_2 and N_2O are released to the atmosphere – **thereby completing the microbial N cycle.**
- **Detrimental process for agriculture**, as fixed N from soil gets removed (in particular in waterlogged soils) – fixed N loss
- N_2O can be photochemically oxidized to NO in the atmosphere - NO reacts with Ozone (O_3) to form NO_2^- , which returns to earth as nitric acid (**acid rain process**) – increase acidity of soils
- **N_2O is a potent greenhouse gas** (>270 times than CO_2 over a 100-year time scale)
- **Useful process in wastewater treatment** - to avoid N influx in water bodies and eventual eutrophication

7. Dissimilatory Nitrate Reduction to Ammonium

- DNRA or nitrate/nitrite ammonification - anaerobic respiration process.
- Microbes use nitrate as an electron acceptor, reducing it to nitrite, and then directly to ammonium

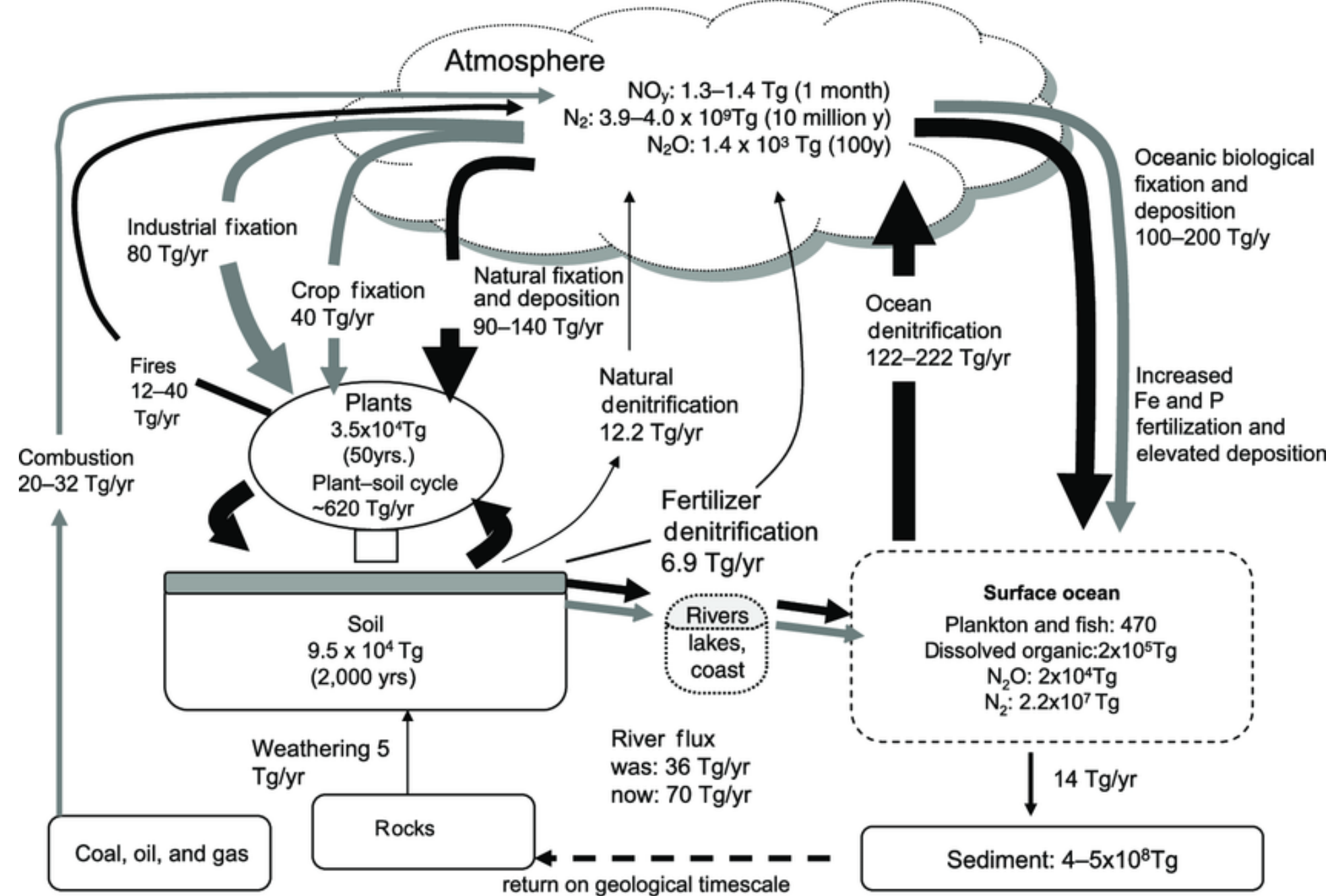
Step 1: Nitrate to nitrite



Step 2: Nitrite to ammonium



- Both denitrifying and DNRA bacteria compete for NO_3^- : DNRA occurs under nitrate limiting conditions – **retains fixed nitrogen in environments**



**Key Nitrogen
pools (Tg),
fluxes (Tg/yr) &
Turnover times
(y – in parentheses)**

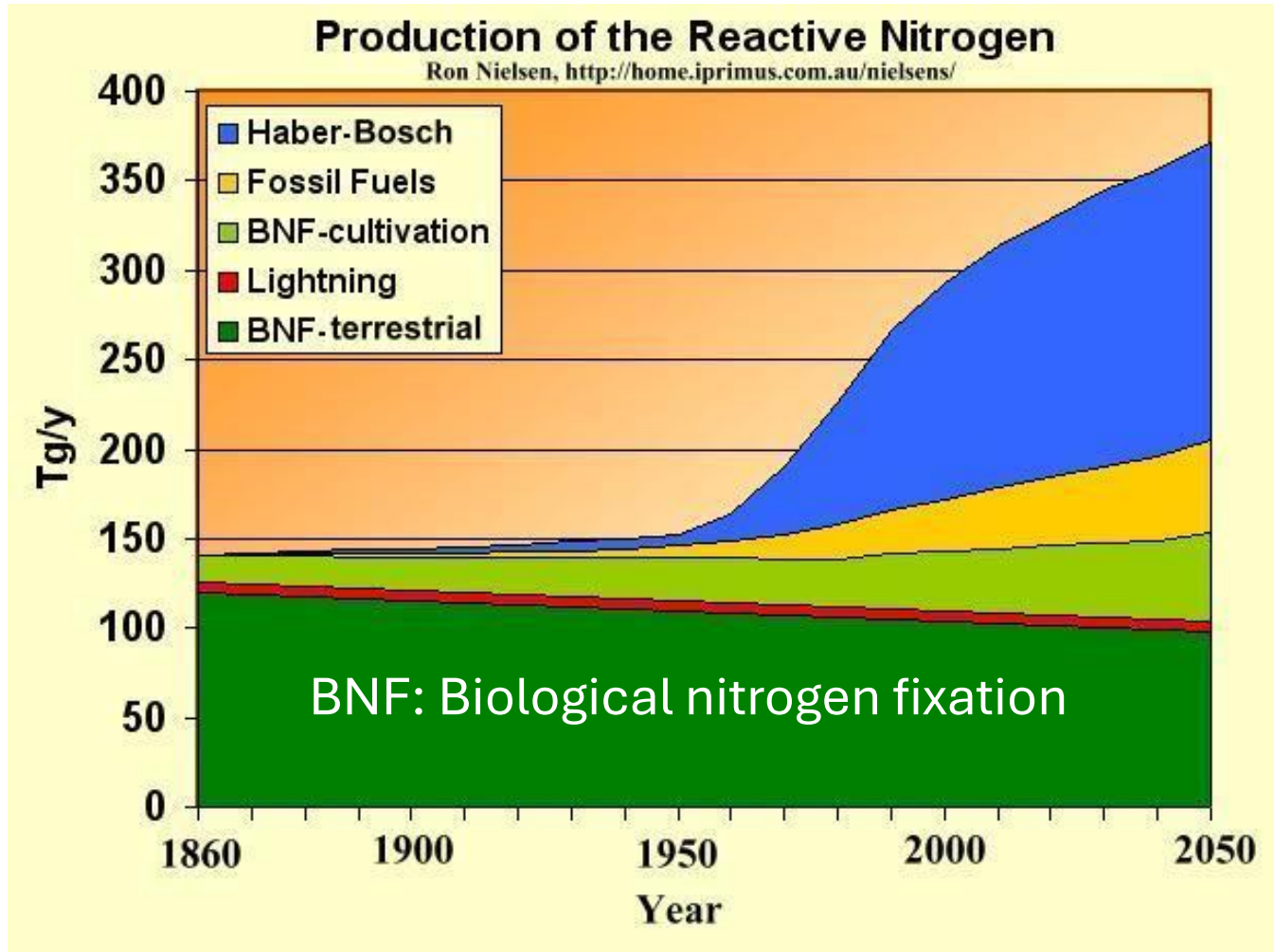
Solid arrows: Pre-industrial cycle

Grey arrows: Human additions

*The width of the arrows is approximately proportional to the flux rates

*To convert Tg N to moles N, multiply by 7.1×10^{10}

Human influence on the global N cycle



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**Doubling of the “N” compounds in the past 100 years
(from ~200 to >400 Tg N/yr)**

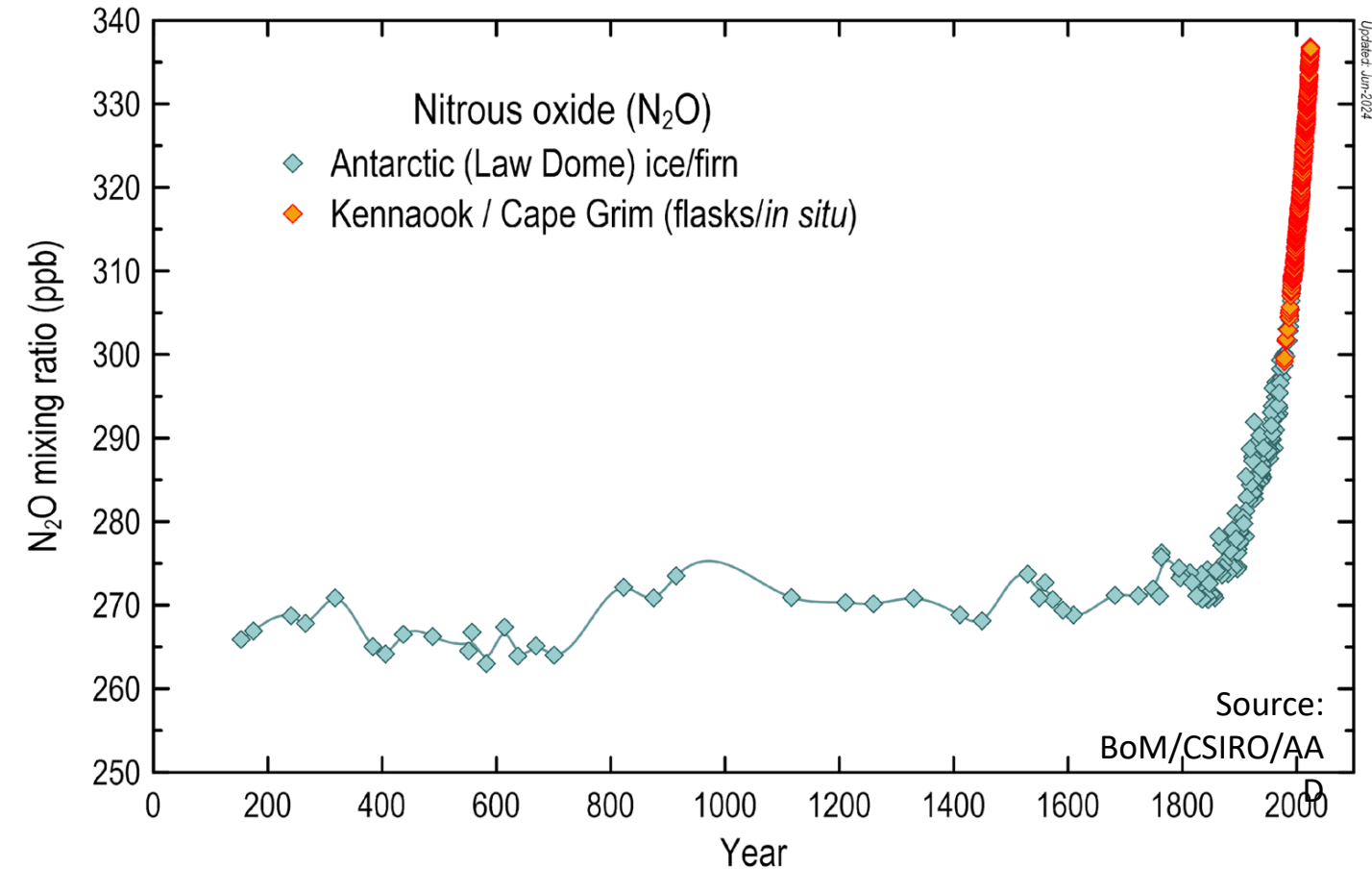
- **Haber-Bosch process:**
nitrogen-based synthetic fertilizers
 - Uses natural gas
 - GHG emissions
- Fossil fuels combustion
- Wastewater discharge
- Agricultural runoff (**> 65 Tg N/Yr**)

Process	Annual Flux
Sources	
Fossil fuel combustion	25
Net emissions from soils	12
Biomass burning	9.6
Lightning	5
NH ₃ oxidation	1
Aircraft	0.4
Transport from the stratosphere	0.6
Total sources	53.6
Sinks	
Deposition on land	24.8
Deposition on the ocean surface	23.0
Total sinks	47.8

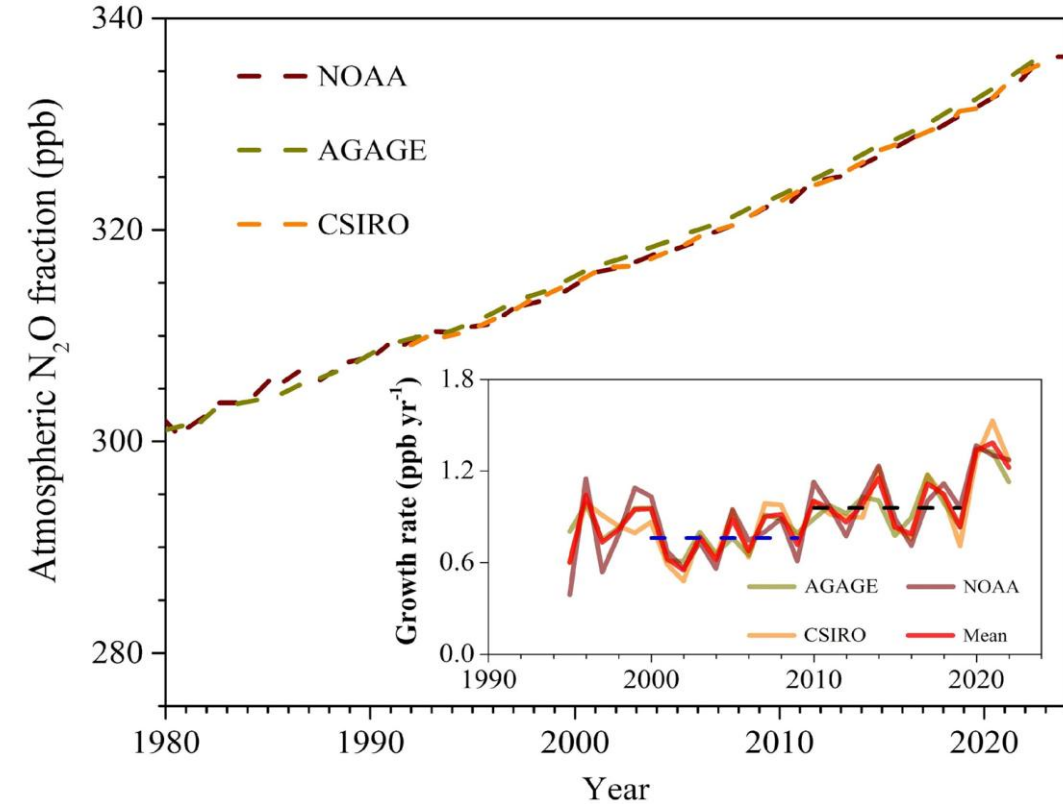
A Global Budget for Atmospheric NO_x (values are Tg N/yr as NO)

imbalance of about 5.5 Tg N/yr

Atmospheric N₂O Concentrations



The global N₂O *concentration* has increased by about 25%, from 270 parts per billion (ppb) in 1750 to 336 ppb in 2022



The mean growth rate for 2010-2019 was 0.96 ppb yr⁻¹.

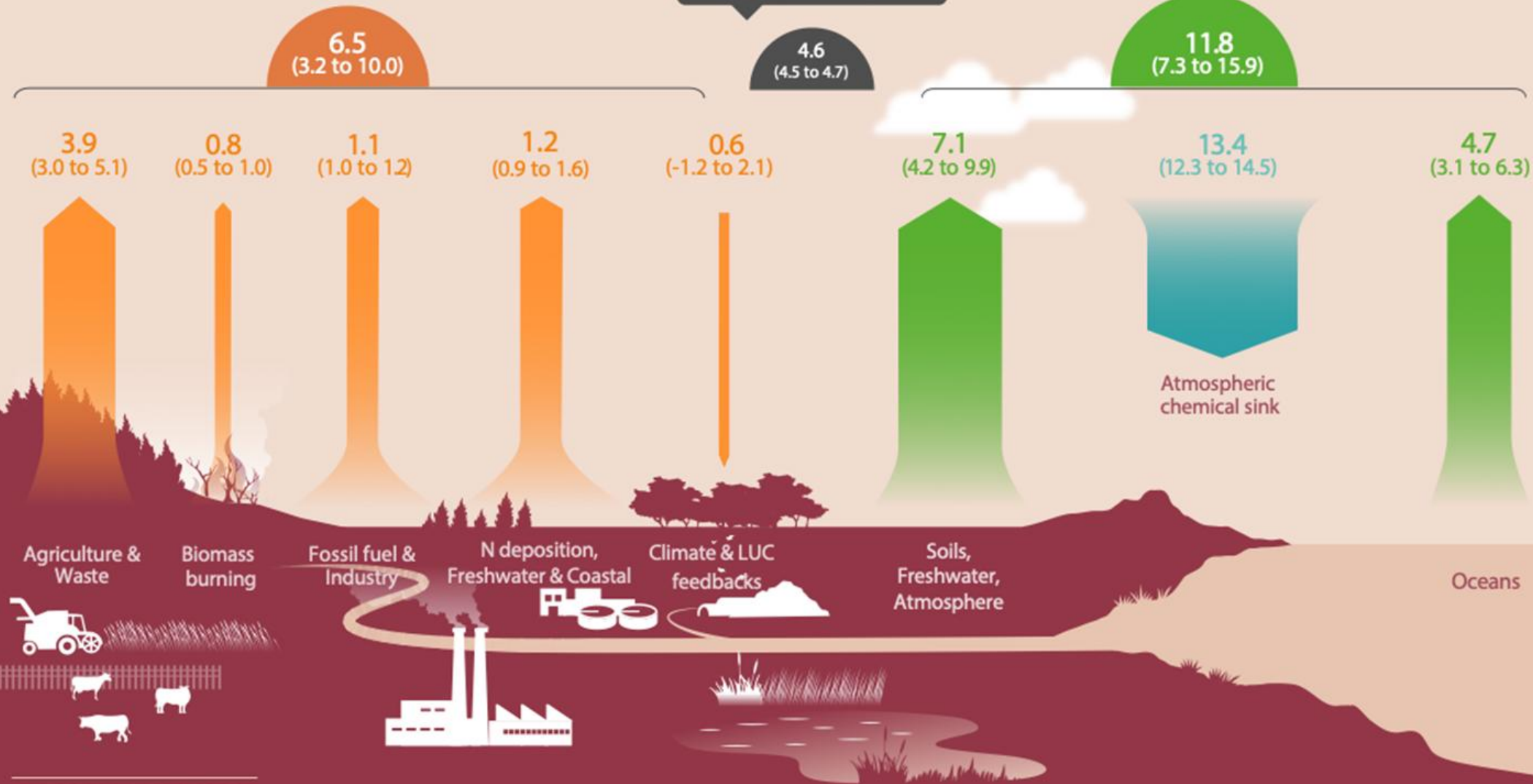
Source: Global Carbon project

GLOBAL N₂O BUDGET

ANTHROPOGENIC SOURCES

CHANGE IN ATMOSPHERIC ABUNDANCE

NATURAL SOURCES



Anthropogenic sources contributed 36% to total global N₂O emissions

FLUX OF N₂O BY SOURCE

in Teragrams of Nitrogen per year (Tg N or million metric tons yr⁻¹) for the decade of 2010-19

Anthropogenic sources/sinks Natural sources Natural sinks

Based on Tian et al. 2024, ESSD

Consequences of increased reactive N in environment

- Impacts on
 - soils (**e.g., acidification**)
 - atmosphere (**e.g., Ozone depletion, greenhouse gas balance**; N_2O 270 ppb in 1750 to 336 ppb in 2022 (25% $\uparrow$))
 - water sources (e.g., **groundwater nitrate contamination** – up to 200 mg/L in Malwa region, Punjab*) and
 - freshwater/seawater (**e.g., eutrophication and dead zones**)
- Affect climate, ecosystems, and human health